



# CHEMISTRY SAMPLE DATA SHEET

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# Periodic table

|                                      |                                       |                                             |                                       |                                      |                                        |                                       |                                       |                                     |                                       |                                    |                                     |                                      |                                       |                                      |                                       |                                      |                                     |
|--------------------------------------|---------------------------------------|---------------------------------------------|---------------------------------------|--------------------------------------|----------------------------------------|---------------------------------------|---------------------------------------|-------------------------------------|---------------------------------------|------------------------------------|-------------------------------------|--------------------------------------|---------------------------------------|--------------------------------------|---------------------------------------|--------------------------------------|-------------------------------------|
| 1                                    | 2                                     | 3                                           | 4                                     | 5                                    | 6                                      | 7                                     | 8                                     | 9                                   | 10                                    | 11                                 | 12                                  | 13                                   | 14                                    | 15                                   | 16                                    | 17                                   | 18                                  |
| 1<br><b>H</b><br>hydrogen<br>1.008   |                                       |                                             |                                       |                                      |                                        |                                       |                                       |                                     |                                       |                                    |                                     |                                      |                                       |                                      |                                       |                                      | 2<br><b>He</b><br>helium<br>4.003   |
| 3<br><b>Li</b><br>lithium<br>6.968   | 4<br><b>Be</b><br>beryllium<br>9.012  |                                             |                                       |                                      |                                        |                                       |                                       |                                     |                                       |                                    |                                     |                                      |                                       |                                      |                                       | 9<br><b>F</b><br>fluorine<br>19.00   | 10<br><b>Ne</b><br>neon<br>20.18    |
| 11<br><b>Na</b><br>sodium<br>22.99   | 12<br><b>Mg</b><br>magnesium<br>24.31 |                                             |                                       |                                      |                                        |                                       |                                       |                                     |                                       |                                    |                                     |                                      |                                       |                                      |                                       | 17<br><b>Cl</b><br>chlorine<br>35.45 | 18<br><b>Ar</b><br>argon<br>39.95   |
| 19<br><b>K</b><br>potassium<br>39.10 | 20<br><b>Ca</b><br>calcium<br>40.08   | 21<br><b>Sc</b><br>scandium<br>44.96        | 22<br><b>Ti</b><br>titanium<br>47.87  | 23<br><b>V</b><br>vanadium<br>50.94  | 24<br><b>Cr</b><br>chromium<br>52.00   | 25<br><b>Mn</b><br>manganese<br>54.94 | 26<br><b>Fe</b><br>iron<br>55.85      | 27<br><b>Co</b><br>cobalt<br>58.93  | 28<br><b>Ni</b><br>nickel<br>58.69    | 29<br><b>Cu</b><br>copper<br>63.55 | 30<br><b>Zn</b><br>zinc<br>65.38    | 31<br><b>Ga</b><br>gallium<br>69.72  | 32<br><b>Ge</b><br>germanium<br>72.63 | 33<br><b>As</b><br>arsenic<br>74.92  | 34<br><b>Se</b><br>selenium<br>78.96  | 35<br><b>Br</b><br>bromine<br>79.90  | 36<br><b>Kr</b><br>krypton<br>83.80 |
| 37<br><b>Rb</b><br>rubidium<br>85.47 | 38<br><b>Sr</b><br>strontium<br>87.62 | 39<br><b>Y</b><br>yttrium<br>88.91          | 40<br><b>Zr</b><br>zirconium<br>91.22 | 41<br><b>Nb</b><br>niobium<br>92.91  | 42<br><b>Mo</b><br>molybdenum<br>95.96 | 43<br><b>Tc</b><br>technetium         | 44<br><b>Ru</b><br>ruthenium<br>101.1 | 45<br><b>Rh</b><br>rhodium<br>102.9 | 46<br><b>Pd</b><br>palladium<br>106.4 | 47<br><b>Ag</b><br>silver<br>107.9 | 48<br><b>Cd</b><br>cadmium<br>112.4 | 49<br><b>In</b><br>indium<br>114.8   | 50<br><b>Sn</b><br>tin<br>118.7       | 51<br><b>Sb</b><br>antimony<br>121.8 | 52<br><b>Te</b><br>tellurium<br>127.6 | 53<br><b>I</b><br>iodine<br>126.9    | 54<br><b>Xe</b><br>xenon<br>131.3   |
| 55<br><b>Cs</b><br>caesium<br>132.9  | 56<br><b>Ba</b><br>barium<br>137.3    | 57-71<br><b>*La</b><br>lanthanum<br>series  | 72<br><b>Hf</b><br>hafnium<br>178.5   | 73<br><b>Ta</b><br>tantalum<br>180.9 | 74<br><b>W</b><br>tungsten<br>183.8    | 75<br><b>Re</b><br>rhenium<br>186.2   | 76<br><b>Os</b><br>osmium<br>190.2    | 77<br><b>Ir</b><br>iridium<br>192.2 | 78<br><b>Pt</b><br>platinum<br>195.1  | 79<br><b>Au</b><br>gold<br>197.0   | 80<br><b>Hg</b><br>mercury<br>200.6 | 81<br><b>Tl</b><br>thallium<br>204.4 | 82<br><b>Pb</b><br>lead<br>207.2      | 83<br><b>Bi</b><br>bismuth<br>209.0  | 84<br><b>Po</b><br>polonium           | 85<br><b>At</b><br>astatine          | 86<br><b>Rn</b><br>radon            |
| 87<br><b>Fr</b><br>francium          | 88<br><b>Ra</b><br>radium             | 89-103<br><b>**Ac</b><br>actinium<br>series | 104<br><b>Rf</b><br>rutherfordium     | 105<br><b>Db</b><br>dubnium          | 106<br><b>Sg</b><br>seaborgium         | 107<br><b>Bh</b><br>bohrium           | 108<br><b>Hs</b><br>hassium           | 109<br><b>Mt</b><br>meitnerium      | 110<br><b>Ds</b><br>darmstadtium      | 111<br><b>Rg</b><br>roentgenium    | 112<br><b>Cn</b><br>copernicium     |                                      | 114<br><b>Fl</b><br>flerovium         |                                      | 116<br><b>Lv</b><br>livermorium       |                                      |                                     |

|           |                   |                                     |                                          |                                       |                               |                                      |                                      |                                        |                                     |                                        |                                     |                                    |                                     |                                       |                                      |
|-----------|-------------------|-------------------------------------|------------------------------------------|---------------------------------------|-------------------------------|--------------------------------------|--------------------------------------|----------------------------------------|-------------------------------------|----------------------------------------|-------------------------------------|------------------------------------|-------------------------------------|---------------------------------------|--------------------------------------|
| <b>*</b>  | Lanthanide series | 58<br><b>Ce</b><br>cerium<br>140.1  | 59<br><b>Pr</b><br>praseodymium<br>140.9 | 60<br><b>Nd</b><br>neodymium<br>144.2 | 61<br><b>Pm</b><br>promethium | 62<br><b>Sm</b><br>samarium<br>150.4 | 63<br><b>Eu</b><br>europium<br>152.0 | 64<br><b>Gd</b><br>gadolinium<br>157.3 | 65<br><b>Tb</b><br>terbium<br>158.9 | 66<br><b>Dy</b><br>dysprosium<br>162.5 | 67<br><b>Ho</b><br>holmium<br>164.9 | 68<br><b>Er</b><br>erbium<br>167.3 | 69<br><b>Tm</b><br>thulium<br>168.9 | 70<br><b>Yb</b><br>ytterbium<br>173.1 | 71<br><b>Lu</b><br>lutetium<br>175.0 |
| <b>**</b> | Actinide series   | 90<br><b>Th</b><br>thorium<br>232.0 | 91<br><b>Pa</b><br>protactinium<br>231.0 | 92<br><b>U</b><br>uranium<br>238.0    | 93<br><b>Np</b><br>neptunium  | 94<br><b>Pu</b><br>plutonium         | 95<br><b>Am</b><br>americium         | 96<br><b>Cm</b><br>curium              | 97<br><b>Bk</b><br>berkelium        | 98<br><b>Cf</b><br>californium         | 99<br><b>Es</b><br>einsteinium      | 100<br><b>Fm</b><br>fermium        | 101<br><b>Md</b><br>mendelevium     | 102<br><b>No</b><br>nobelium          | 103<br><b>Lr</b><br>lawrencium       |

Key:

|               |               |      |                        |
|---------------|---------------|------|------------------------|
| Atomic number | <b>Symbol</b> | Name | Standard atomic weight |
|---------------|---------------|------|------------------------|

[Data source: The International Union of Pure and Applied Chemistry Periodic Table of the Elements (May 2013)]

## Formulae

$$\text{Number of moles} \quad n = \frac{m}{M} = \frac{\text{mass}}{\text{molar mass}}$$

$$\text{Number of moles of solute} \quad n = cV$$

$$\text{Number of moles of a gas at STP} \quad n = \frac{V}{22.71}$$

$$\text{Parts per million} \quad \text{ppm} = \frac{\text{mass of solute (mg)}}{\text{mass of solution (kg)}}$$

$$\text{pH of a solution} \quad \text{pH} = -\log [\text{H}^+]$$

## Units

Volumes are given in the units of litres (L), or millilitres (mL)

Temperatures are given in the units of degrees Celsius (°C) or kelvin (K).

It may be assumed that 0.0 °C = 273.15 K

Energy changes are given in kilojoules (kJ)

Pressures are given in kilopascals (kPa)

Solution concentrations are given in the units moles per litre (mol L<sup>-1</sup>), grams per litre (g L<sup>-1</sup>) or parts per million (ppm).

## Constants

Universal gas constant, R = 8.314 J K<sup>-1</sup> mol<sup>-1</sup>

Avogadro constant, N = 6.022 × 10<sup>23</sup> mol<sup>-1</sup>

Volume of 1.00 mol of an ideal gas at 0.0 °C and 100.0 kPa is 22.71 L

S.T.P. is 0.0 °C and 100.0 kPa

Equilibrium constant for water at 25 °C, K<sub>w</sub> = 1.00 × 10<sup>-14</sup>

## Solubility rules for ionic solids in water

### Soluble in water

| Soluble        | Exceptions                                                |                                                     |
|----------------|-----------------------------------------------------------|-----------------------------------------------------|
|                | Insoluble                                                 | Slightly soluble                                    |
| Most chlorides | AgCl                                                      | PbCl <sub>2</sub>                                   |
| Most bromides  | AgBr                                                      | PbBr <sub>2</sub>                                   |
| Most iodides   | AgI, PbI <sub>2</sub>                                     |                                                     |
| All nitrates   | No exceptions                                             |                                                     |
| All ethanoates |                                                           |                                                     |
| Most sulfates  | SrSO <sub>4</sub> , BaSO <sub>4</sub> , PbSO <sub>4</sub> | CaSO <sub>4</sub> , Ag <sub>2</sub> SO <sub>4</sub> |

### Insoluble in water

| Insoluble       | Exceptions                                                                                                         |                                           |
|-----------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
|                 | Soluble                                                                                                            | Slightly soluble                          |
| Most hydroxides | NaOH, KOH, Ba(OH) <sub>2</sub><br>NH <sub>4</sub> OH*, AgOH**                                                      | Ca(OH) <sub>2</sub> , Sr(OH) <sub>2</sub> |
| Most carbonates | Na <sub>2</sub> CO <sub>3</sub> , K <sub>2</sub> CO <sub>3</sub> , (NH <sub>4</sub> ) <sub>2</sub> CO <sub>3</sub> |                                           |
| Most phosphates | Na <sub>3</sub> PO <sub>4</sub> , K <sub>3</sub> PO <sub>4</sub> , (NH <sub>4</sub> ) <sub>3</sub> PO <sub>4</sub> |                                           |
| Most sulfides   | Na <sub>2</sub> S, K <sub>2</sub> S, (NH <sub>4</sub> ) <sub>2</sub> S                                             |                                           |

\* NH<sub>3</sub> dissolves in water to form both NH<sub>3</sub> (aq) and NH<sub>4</sub><sup>+</sup>(aq)/OH<sup>-</sup>(aq)

\*\* Ag<sup>+</sup>(aq) reacts with OH<sup>-</sup>(aq) to form insoluble Ag<sub>2</sub>O

Soluble = more than 0.1 mole dissolves per litre

Slightly soluble = between 0.01 and 0.1 mole dissolves per litre

Insoluble = less than 0.01 mole dissolves per litre

## Colours of selected substances

In general, ionic solids have the same colour as that of any coloured ion they contain.

Two colourless ions in general produce a white solid.

Selected exceptions to these two basic rules are noted below.

| Ionic Solid          | Colour      |
|----------------------|-------------|
| copper(II) carbonate | green       |
| copper(II) chloride  | green       |
| copper(II) oxide     | black       |
| copper(II) sulfide   | black       |
| lead(II) iodide      | yellow      |
| lead(II) sulfide     | grey        |
| manganese(IV) oxide  | black       |
| silver carbonate     | yellow      |
| silver iodide        | pale yellow |
| silver oxide         | brown       |
| silver sulfide       | black       |

### Other coloured substances

Most gases and liquids are colourless, and most metals are silvery or grey. Selected exceptions to these basic rules are noted below.

| Substance        | State | Colour      |
|------------------|-------|-------------|
| copper           | solid | salmon pink |
| gold             | solid | yellow      |
| nitrogen dioxide | gas   | brown       |
| sulfur           | solid | yellow      |

### Coloured halogens

| Halogen             | Colour of free element |
|---------------------|------------------------|
| F <sub>2</sub> (g)  | yellow                 |
| Cl <sub>2</sub> (g) | greenish-yellow        |
| Br <sub>2</sub> (l) | red                    |
| I <sub>2</sub> (s)  | purple                 |

| Halogen              | Colour of halogen in aqueous solution |
|----------------------|---------------------------------------|
| Cl <sub>2</sub> (aq) | pale yellow                           |
| Br <sub>2</sub> (aq) | orange                                |
| I <sub>2</sub> (aq)  | brown                                 |

| Halogen         | Colour of halogen in organic solvent |
|-----------------|--------------------------------------|
| Br <sub>2</sub> | red                                  |
| I <sub>2</sub>  | purple                               |

### Coloured ions in aqueous solution

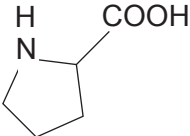
| Cation           | Colour     |
|------------------|------------|
| Cr <sup>3+</sup> | deep green |
| Co <sup>2+</sup> | pink       |
| Cu <sup>2+</sup> | blue       |
| Fe <sup>2+</sup> | pale green |
| Fe <sup>3+</sup> | pale brown |
| Mn <sup>2+</sup> | pale pink  |
| Ni <sup>2+</sup> | green      |

| Anion                                        | Colour |
|----------------------------------------------|--------|
| CrO <sub>4</sub> <sup>2-</sup>               | yellow |
| Cr <sub>2</sub> O <sub>7</sub> <sup>2-</sup> | orange |
| MnO <sub>4</sub> <sup>-</sup>                | purple |

## $\alpha$ -amino acids

| Name          | Symbol | Structure                                                                                                                                                                    |
|---------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alanine       | Ala    | $\begin{array}{c} \text{CH}_3 \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$                                                                              |
| arginine      | Arg    | $\begin{array}{c} \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{NH} - \text{C}(=\text{NH}) - \text{NH}_2 \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$ |
| asparagine    | Asn    | $\begin{array}{c} \text{O} \\    \\ \text{CH}_2 - \text{C} - \text{NH}_2 \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$                                   |
| aspartic acid | Asp    | $\begin{array}{c} \text{CH}_2 - \text{COOH} \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$                                                                |
| cysteine      | Cys    | $\begin{array}{c} \text{CH}_2 - \text{SH} \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$                                                                  |
| glutamine     | Gln    | $\begin{array}{c} \text{O} \\    \\ \text{CH}_2 - \text{CH}_2 - \text{C} - \text{NH}_2 \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$                     |
| glutamic acid | Glu    | $\begin{array}{c} \text{CH}_2 - \text{CH}_2 - \text{COOH} \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$                                                  |
| glycine       | Gly    | $\text{H}_2\text{N} - \text{CH}_2 - \text{COOH}$                                                                                                                             |
| histidine     | His    | $\begin{array}{c} \text{N} \\ // \quad \backslash \\ \text{CH}_2 - \text{C} \quad \text{N} - \text{H} \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$      |
| isoleucine    | Ile    | $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH}_2 - \text{CH}_3 \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$                                      |

## $\alpha$ -amino acids

| Name          | Symbol | Structure                                                                                                                                               |
|---------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| leucine       | Leu    | $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH}_3 \\   \\ \text{CH}_2 \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$           |
| lysine        | Lys    | $\begin{array}{c} \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{NH}_2 \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$ |
| methionine    | Met    | $\begin{array}{c} \text{CH}_2 - \text{CH}_2 - \text{S} - \text{CH}_3 \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$                  |
| phenylalanine | Phe    | $\begin{array}{c} \text{CH}_2 - \text{C}_6\text{H}_5 \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$                                  |
| proline       | Pro    |                                                                      |
| serine        | Ser    | $\begin{array}{c} \text{CH}_2 - \text{OH} \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$                                             |
| threonine     | Thr    | $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{OH} \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$                                 |
| tryptophan    | Trp    | $\begin{array}{c} \text{CH}_2 - \text{C}_8\text{H}_6\text{N}_2 \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$                        |
| tyrosine      | Tyr    | $\begin{array}{c} \text{CH}_2 - \text{C}_6\text{H}_4 - \text{OH} \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$                      |
| valine        | Val    | $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH}_3 \\   \\ \text{H}_2\text{N} - \text{CH} - \text{COOH} \end{array}$                               |

## Standard Reduction Potentials at 25 °C

| Half-reaction                                                                                                                                                           | E°(volts) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| $\text{F}_2(\text{g}) + 2 \text{e}^- \rightleftharpoons 2 \text{F}^-(\text{aq})$                                                                                        | + 2.89    |
| $\text{H}_2\text{O}_2(\text{aq}) + 2 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightleftharpoons 2 \text{H}_2\text{O}(\ell)$                                                | + 1.76    |
| $\text{PbO}_2(\text{s}) + \text{SO}_4^{2-}(\text{aq}) + 4 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{PbSO}_4(\text{s}) + 2 \text{H}_2\text{O}(\ell)$ | + 1.69    |
| $2 \text{HClO}(\text{aq}) + 2 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Cl}_2(\text{g}) + 2 \text{H}_2\text{O}(\ell)$                               | + 1.63    |
| $\text{MnO}_4^-(\text{aq}) + 8 \text{H}^+(\text{aq}) + 5 \text{e}^- \rightleftharpoons \text{Mn}^{2+}(\text{aq}) + 4 \text{H}_2\text{O}(\ell)$                          | + 1.51    |
| $\text{Au}^{3+}(\text{aq}) + 3 \text{e}^- \rightleftharpoons \text{Au}(\text{s})$                                                                                       | + 1.50    |
| $\text{HClO}(\text{aq}) + \text{H}^+(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\ell)$                                    | + 1.49    |
| $\text{PbO}_2(\text{s}) + 4 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Pb}^{2+}(\text{aq}) + 2 \text{H}_2\text{O}(\ell)$                             | + 1.46    |
| $\text{Cl}_2(\text{g}) + 2 \text{e}^- \rightleftharpoons 2 \text{Cl}^-(\text{aq})$                                                                                      | + 1.36    |
| $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14 \text{H}^+(\text{aq}) + 6 \text{e}^- \rightleftharpoons 2 \text{Cr}^{3+}(\text{aq}) + 7 \text{H}_2\text{O}(\ell)$           | + 1.36    |
| $\text{O}_2(\text{g}) + 4 \text{H}^+(\text{aq}) + 4 \text{e}^- \rightleftharpoons 2 \text{H}_2\text{O}(\ell)$                                                           | + 1.23    |
| $\text{Br}_2(\ell) + 2 \text{e}^- \rightleftharpoons 2 \text{Br}^-(\text{aq})$                                                                                          | + 1.08    |
| $\text{Ag}^+(\text{aq}) + \text{e}^- \rightleftharpoons \text{Ag}(\text{s})$                                                                                            | + 0.80    |
| $\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightleftharpoons \text{Fe}^{2+}(\text{aq})$                                                                                   | + 0.77    |
| $\text{O}_2(\text{g}) + 2 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{H}_2\text{O}_2(\text{aq})$                                                      | + 0.70    |
| $\text{I}_2(\text{s}) + 2 \text{e}^- \rightleftharpoons 2 \text{I}^-(\text{aq})$                                                                                        | + 0.54    |
| $\text{O}_2(\text{g}) + 2 \text{H}_2\text{O}(\ell) + 4 \text{e}^- \rightleftharpoons 4 \text{OH}^-(\text{aq})$                                                          | + 0.40    |
| $\text{Cu}^{2+}(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Cu}(\text{s})$                                                                                       | + 0.34    |
| $\text{S}(\text{s}) + 2 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{H}_2\text{S}(\text{aq})$                                                          | + 0.17    |
| $2 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{H}_2(\text{g})$                                                                                        | 0 exactly |
| $\text{Pb}^{2+}(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Pb}(\text{s})$                                                                                       | – 0.13    |
| $\text{Sn}^{2+}(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Sn}(\text{s})$                                                                                       | – 0.14    |
| $\text{Ni}^{2+}(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Ni}(\text{s})$                                                                                       | – 0.24    |
| $\text{Co}^{2+}(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Co}(\text{s})$                                                                                       | – 0.28    |
| $\text{PbSO}_4(\text{s}) + 2 \text{e}^- \rightleftharpoons \text{Pb}(\text{s}) + \text{SO}_4^{2-}(\text{aq})$                                                           | – 0.36    |
| $\text{Cd}^{2+}(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Cd}(\text{s})$                                                                                       | – 0.40    |
| $2 \text{CO}_2(\text{g}) + 2 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{H}_2\text{C}_2\text{O}_4(\text{aq})$                                         | – 0.43    |
| $\text{Fe}^{2+}(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Fe}(\text{s})$                                                                                       | – 0.44    |
| $\text{Cr}^{3+}(\text{aq}) + 3 \text{e}^- \rightleftharpoons \text{Cr}(\text{s})$                                                                                       | – 0.74    |
| $\text{Zn}^{2+}(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Zn}(\text{s})$                                                                                       | – 0.76    |
| $2 \text{H}_2\text{O}(\ell) + 2 \text{e}^- \rightleftharpoons \text{H}_2(\text{g}) + 2 \text{OH}^-(\text{aq})$                                                          | – 0.83    |
| $\text{Mn}^{2+}(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Mn}(\text{s})$                                                                                       | – 1.18    |
| $\text{Al}^{3+}(\text{aq}) + 3 \text{e}^- \rightleftharpoons \text{Al}(\text{s})$                                                                                       | – 1.68    |
| $\text{Mg}^{2+}(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Mg}(\text{s})$                                                                                       | – 2.36    |
| $\text{Na}^+(\text{aq}) + \text{e}^- \rightleftharpoons \text{Na}(\text{s})$                                                                                            | – 2.71    |
| $\text{Ca}^{2+}(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Ca}(\text{s})$                                                                                       | – 2.87    |
| $\text{Sr}^{2+}(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Sr}(\text{s})$                                                                                       | – 2.90    |
| $\text{Ba}^{2+}(\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Ba}(\text{s})$                                                                                       | – 2.91    |
| $\text{K}^+(\text{aq}) + \text{e}^- \rightleftharpoons \text{K}(\text{s})$                                                                                              | – 2.94    |

[Data source: Aylward, G.H., & Findlay, T. (2008). *SI Chemical Data* (6th ed.). Queensland: John Wiley & Sons Australia, Ltd.]